Lecture 12: Section 3.4

Related Rates

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What are Related Rates?

Definition

Related rates problems involve finding the rate at which some quantity is changing by relating the quantity to other quantities whose rates of change are known.

The Big Idea

If two or more quantities are related by an equation, then their rates of change (derivatives) are also related.

Example

- How fast is the water level rising in a tank?
- How fast is a shadow length changing?
- How fast is the area of a spill increasing?

Strategy for Solving Related Rates Problems

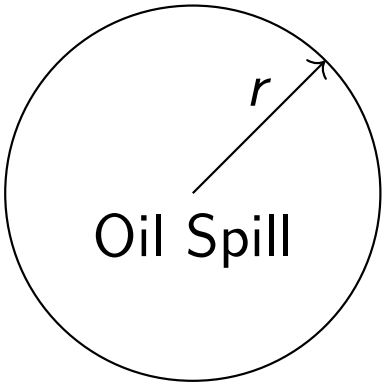
Step-by-Step Approach

1. **Read** the problem carefully
2. **Draw** a diagram if possible
3. **Introduce** notation for variables.
   Assign letters to all quantities that vary with time and any others that seem relevant to the problem. Give a definition for each letter.
4. **Write** an equation relating the variables.
   Identify the rates of change that are known and the rate of change that is to be found. Interpret each rate as a derivative.
5. **Differentiate** both sides with respect to time t
6. **Substitute** known values and rates
7. **Solve** for the unknown rate
8. **Interpret** the answer

Problem 1: Expanding Circle

Oil spilled from a tanker spreads in a circular pattern whose radius increases at a constant rate of 2 ft/s. How fast is the area of the spill increasing when the radius is 60 ft?

Solution 1: Expanding Circle



1. Known: $\frac{dr}{dt} = 2$ ft/s, $r = 60$ ft
2. Area formula: $A = \pi r^2$
3. Differentiate: $\frac{dA}{dt} = 2\pi r \frac{dr}{dt}$
4. Substitute: $\frac{dA}{dt} = 2\pi(60)(2)$
5. Calculate: $\frac{dA}{dt} = 240\pi \approx 754$ ft^2/s

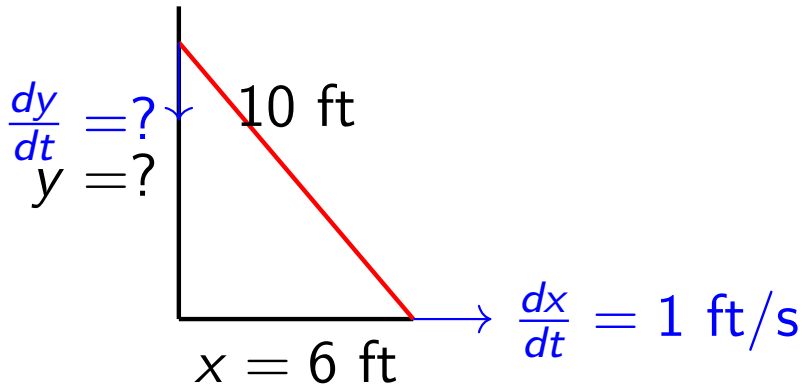
Interpretation

When the radius is 60 ft, the area is increasing at about 754 square feet per second.

Problem 2: Ladder Sliding Down Wall

A 10-ft ladder leans against a wall. The bottom is pulled away at 1 ft/s. How fast is the top sliding down when the bottom is 6 ft from the wall?

Solution 2: Ladder Sliding Down Wall



1. Pythagorean theorem: $x^2 + y^2 = 100$
2. When $x = 6$: $36 + y^2 = 100 \Rightarrow y = 8$ ft
3. Differentiate: $2x\frac{dx}{dt} + 2y\frac{dy}{dt} = 0$
4. Substitute: $2(6)(1) + 2(8)\frac{dy}{dt} = 0$
5. Solve: $16\frac{dy}{dt} = -12 \Rightarrow \frac{dy}{dt} = -0.75$ ft/s

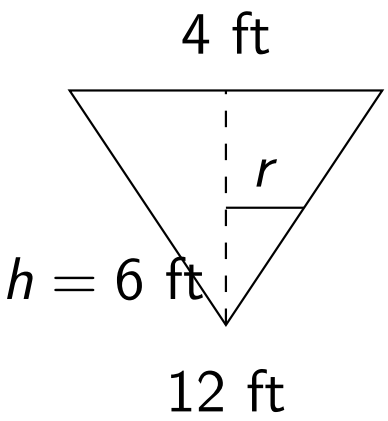
Interpretation

The top is sliding down at 0.75 ft/s (negative indicates downward motion).

Problem 3: Conical Tank

Water drains from a conical tank (vertex down) with height 12 ft and radius 4 ft. The water level drops at 0.5 ft/min. How fast is the water volume decreasing when the water depth is 6 ft?

Solution 3: Conical Tank



1. Known: $\frac{dh}{dt} = -0.5$ ft/min, $h = 6$ ft
2. Similar triangles: $\frac{r}{h} = \frac{4}{12} \Rightarrow r = \frac{h}{3}$
3. Volume:
   $V = \frac{1}{3}\pi r^2 h = \frac{1}{3}\pi \left(\frac{h}{3}\right)^2 h = \frac{\pi}{27}h^3$
4. Differentiate: $\frac{dV}{dt} = \frac{\pi}{9}h^2\frac{dh}{dt}$
5. Substitute: $\frac{dV}{dt} = \frac{\pi}{9}(36)(-0.5)$
6. Calculate: $\frac{dV}{dt} = -2\pi \approx -6.28$ ft^3/min

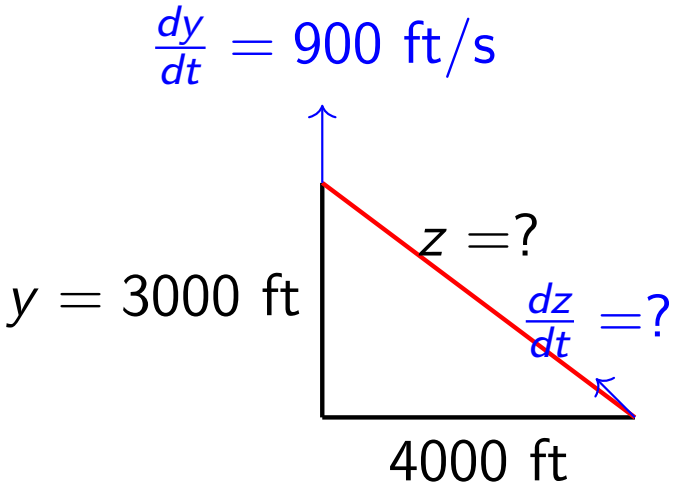
Interpretation

When depth is 6 ft, volume is decreasing at about 6.28 cubic feet per minute.

Problem 4: Rocket Launch

A camera is 4000 ft from a rocket launch pad. The rocket rises vertically at 900 ft/s. How fast is the camera-to-rocket distance changing when the rocket is 3000 ft high?

Solution 4: Rocket Launch



1. Pythagorean theorem: $z^2 = 4000^2 + y^2$
2. When $y = 3000$:
   $z^2 = 16{,}000{,}000 + 9{,}000{,}000 = 25{,}000{,}000 \Rightarrow z = 5000$ ft
3. Differentiate: $2z\frac{dz}{dt} = 2y\frac{dy}{dt}$
4. Substitute: $2(5000)\frac{dz}{dt} = 2(3000)(900)$
5. Solve:
   $10000\frac{dz}{dt} = 5{,}400{,}000 \Rightarrow \frac{dz}{dt} = 540$ ft/s

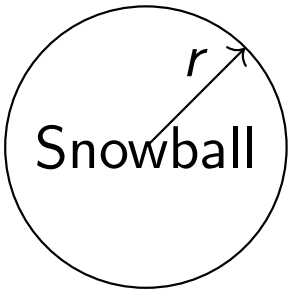
Interpretation

When rocket is 3000 ft high, the camera-to-rocket distance increases at 540 ft/s.

Problem 5: Melting Snowball

A spherical snowball melts so that its surface area decreases at 3 cm^2/min. How fast is the diameter decreasing when the diameter is 10 cm?

Solution 5: Melting Snowball



1. Known: $\frac{dA}{dt} = -3 \text{ cm}^2/\text{min}$, $d = 10$ cm
2. Surface area: $A = 4\pi r^2 = \pi d^2$
3. Differentiate: $\frac{dA}{dt} = 2\pi d \frac{dd}{dt}$
4. Substitute: $-3 = 2\pi(10)\frac{dd}{dt}$
5. Solve: $\frac{dd}{dt} = \frac{-3}{20\pi} \approx -0.0477$ cm/min

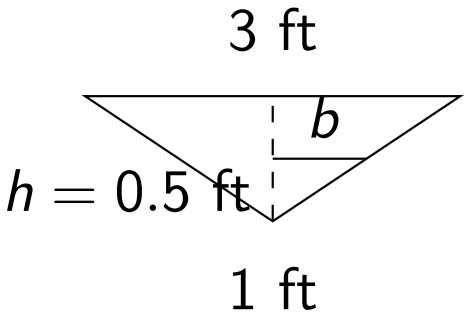
Interpretation

When diameter is 10 cm, it's decreasing at about 0.048 cm per minute.

Problem 6: Filling a Trough

A water trough is 10 ft long with isosceles triangular ends (base 3 ft, height 1 ft). Water flows in at 12 ft^3/min. How fast is the water level rising when the water is 0.5 ft deep?

Solution 6: Filling a Trough



1. Known: $\frac{dV}{dt} = 12$ ft^3/min, $h = 0.5$ ft
2. Similar triangles: $\frac{b}{h} = \frac{3}{1} \Rightarrow b = 3h$
3. Cross-sectional area: $A = \frac{1}{2}bh = \frac{1}{2}(3h)h = 1.5h^2$
4. Volume: $V = 10A = 15h^2$
5. Differentiate: $\frac{dV}{dt} = 30h\frac{dh}{dt}$
6. Substitute: $12 = 30(0.5)\frac{dh}{dt}$
7. Solve: $\frac{dh}{dt} = \frac{12}{15} = 0.8$ ft/min

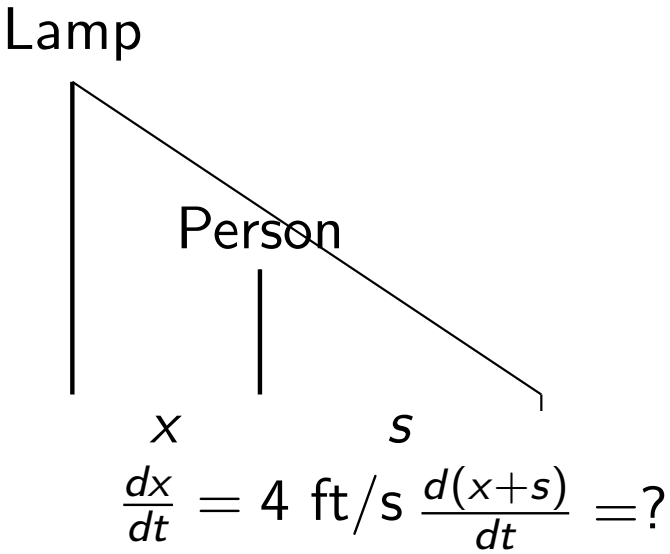
Interpretation

When water is 0.5 ft deep, the level rises at 0.8 ft/min.

Problem 7: Moving Shadow

A 6-ft tall person walks away from a 15-ft lamp post at 4 ft/s. How fast is the tip of their shadow moving?

Solution 7: Moving Shadow



1. Similar triangles: $\frac{15}{x+s} = \frac{6}{s}$
2. Cross multiply:
   $15s = 6x + 6s \Rightarrow 9s = 6x \Rightarrow s = \frac{2}{3}x$
3. Shadow tip position:
   $x + s = x + \frac{2}{3}x = \frac{5}{3}x$
4. Differentiate: $\frac{d}{dt}(x+s) = \frac{5}{3}\frac{dx}{dt}$
5. Substitute: $\frac{d}{dt}(x+s) = \frac{5}{3}(4) = \frac{20}{3}$ ft/s

Interpretation

The shadow tip moves at $\frac{20}{3} \approx 6.67$ ft/s, regardless of position!

Practice Problems

1. A spherical balloon inflates at 100 cm^3/s. How fast is the radius increasing when radius is 25 cm?
2. A 13-ft ladder slides down a wall. The bottom moves at 2 ft/s. How fast is the top moving when bottom is 5 ft from the wall?
3. Sand pours onto a pile at 50 ft^3/min, forming a cone with height equal to base radius. How fast is height increasing when pile is 10 ft high?
4. A plane flies horizontally at 500 mph, 1 mile high. How fast is the distance to observer increasing when plane is 2 miles from observer?
5. Water flows into cylindrical tank (radius 4 ft) at 20 ft^3/min. How fast is water level rising?
6. A man 6 ft tall walks toward a light 15 ft high at 3 ft/s. How fast is shadow length changing?
7. The radius of a circle increases at 0.5 cm/s. How fast is area increasing when circumference is 20π cm?
8. A rectangular trough is 12 ft long, 4 ft wide. Water flows in at 8 ft^3/min. How fast is depth increasing?

Practice Problem Solutions

1. $\frac{dr}{dt} = \frac{1}{25\pi}$ cm/s
2. $\frac{dy}{dt} = -\frac{5}{6}$ ft/s
3. $\frac{dh}{dt} = \frac{1}{2\pi}$ ft/min
4. $\frac{dd}{dt} = 200\sqrt{5} \approx 447$ mph
5. $\frac{dh}{dt} = \frac{5}{4\pi}$ ft/min
6. $\frac{ds}{dt} = -2$ ft/s
7. $\frac{dA}{dt} = 20\pi$ cm^2/s
8. $\frac{dh}{dt} = \frac{1}{6}$ ft/min

Common Related Rates Formulas

Geometry Formulas

- Circle: $A = \pi r^2$, $C = 2\pi r$
- Sphere: $V = \frac{4}{3}\pi r^3$, $A = 4\pi r^2$
- Cylinder: $V = \pi r^2 h$
- Cone: $V = \frac{1}{3}\pi r^2 h$
- Right triangle: $a^2 + b^2 = c^2$

Key Relationships

- Similar triangles for proportional relationships
- Pythagorean theorem for distances
- Volume formulas for filling/emptying problems
- Trigonometric functions for angles

Tips for Success

Common Mistakes to Avoid

- Forgetting the chain rule when differentiating
- Substituting values before differentiating
- Mixing up given rates and unknown rates
- Not drawing a clear diagram
- Forgetting units in final answer

Problem-Solving Strategy

1. Read carefully and identify all rates
2. Draw and label a diagram
3. Write the relating equation
4. Differentiate with respect to time
5. Substitute known values
6. Solve and interpret